

WEEK 1: REQUIREMENT ANALYSIS AND CUSTOMER DATABASE MODULE

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1. Requirement Analysis:

UNDERSTANDING BUSINESS PROBLEM

The Customized Phone Case Ordering System is an e-commerce application that enables customers to design and purchase personalized phone cases online. Customers can choose their phone model, upload their own images, add custom text, select colors and design options, preview the final product and place orders. The system also helps the business manage products, customer information, orders, payments, and deliveries efficiently.

2. Process Flow:



3. Process Flow Description – Customized Phone Case Ordering System

- **Browse / Search Products:** The customer searches for the required phone case model from the available products.
- **Add to Cart:** The customer selects the desired phone case and adds it to the shopping cart.
- **Customize:** The customer uploads an image, adds custom text, and selects design options such as font, color, and layout.
- **Preview:** The customer views the customized phone case design and confirms the final design.
- **Checkout / Payment:** The customer provides the delivery address and completes the payment using an available payment method.
- **Order Placed:** The system generates an order ID and confirms the order.
- **Production:** The customized phone case is prepared according to the customer's selected design.
- **Delivery:** The order is dispatched, and the delivery status and tracking details are updated.
- **Order Delivered:** The customer receives the customized phone case and the order is marked as delivered.

- **View Order History:** The customer can view previous orders and their order status.
- **Admin Management:** The admin manages products, customers, orders, payments, and delivery information.

4. Entities:

Entity	Description	Key Attributes	Key Relationships
Customer	End-user who browses, customizes, and purchases phone cases.	Customer_ID (PK), First_Name, Last_Name, Email, Phone_Number, Password, Address	One-to-Many with Order; One-to-One with Cart; One-to-Many with Customization
Product	Phone case available for customers to purchase and customize.	Product_ID (PK), Product_Name, Brand, Phone_Model, Material, Price, Stock_Quantity	One-to-Many with Order_Item; One-to-Many with Customization
Customization	Stores the design details selected by the customer for a phone case.	Customization_ID (PK), Customer_ID (FK), Product_ID (FK), Uploaded_Image, Custom_Text, Font_Style, Font_Color, Background_Color	Many-to-One with Customer; Many-to-One with Product; One-to-Many with Order_Item
Cart	Stores the phone cases selected by a customer before placing an order.	Cart_ID (PK), Customer_ID (FK), Total_Amount, Created_Date	One-to-One with Customer; Many-to-Many with Product
Order	Represents a confirmed purchase made by a customer.	Order_ID (PK), Customer_ID (FK), Order_Date, Order_Status, Total_Amount	Many-to-One with Customer; One-to-Many with Order_Item; One-to-One with Payment; One-to-One with Delivery

Order_Item	Represents individual phone cases included in an order.	Order_Item_ID (PK), Order_ID (FK), Product_ID (FK), Customization_ID (FK), Quantity, Price, Subtotal	Many-to-One with Order; Many-to-One with Product; Many-to-One with Customization
Payment	Stores payment information related to an order.	Payment_ID (PK), Order_ID (FK), Payment_Method, Payment_Status, Payment_Date, Amount	One-to-One with Order
Delivery	Stores shipping and delivery information for an order.	Delivery_ID (PK), Order_ID (FK), Delivery_Address, Courier_Name, Tracking_Number, Delivery_Status, Delivery_Date	One-to-One with Order
Admin	Manages products, customers, orders, payments, and delivery information.	Admin_ID (PK), Admin_Name, Email, Password, Role	One-to-Many with Product; One-to-Many with Order

5. Customer Table creation:

```
SQL> CREATE TABLE Customer(Customer_ID varchar(5) Primary key, Customer_Name varchar(30) not null, Email varchar(30) not null, Phone varchar(10), Address varchar(200), Registration_Date date);
Table created.
```

```
SQL> desc Customer;
+-----+-----+-----+
Name          Null?   Type
+-----+-----+-----+
CUSTOMER_ID    NOT NULL VARCHAR2(5)
CUSTOMER_NAME  NOT NULL VARCHAR2(30)
EMAIL          NOT NULL VARCHAR2(30)
PHONE         NOT NULL VARCHAR2(10)
ADDRESS        NOT NULL VARCHAR2(200)
REGISTRATION_DATE NOT NULL DATE
```

6. Inserting Values:

```
SQL> INSERT INTO Customer
  2 (CUSTOMER_ID, CUSTOMER_NAME, EMAIL, PHONE, ADDRESS, REGISTRATION_DATE)
  3 VALUES
  4 ('C001', 'Arun Kumar', 'arun@gmail.com', '9876543210', 'Chennai', TO_DATE('10-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO Customer
  2 (CUSTOMER_ID, CUSTOMER_NAME, EMAIL, PHONE, ADDRESS, REGISTRATION_DATE)
  3 VALUES
  4 ('C002', 'Priya Sharma', 'priya@gmail.com', '9876543211', 'Tambaram', TO_DATE('11-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO Customer
  2 (CUSTOMER_ID, CUSTOMER_NAME, EMAIL, PHONE, ADDRESS, REGISTRATION_DATE)
  3 VALUES
  4 ('C003', 'Rahul Raj', 'rahul@gmail.com', '9876543212', 'Chengalpattu', TO_DATE('12-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO Customer
  2 (CUSTOMER_ID, CUSTOMER_NAME, EMAIL, PHONE, ADDRESS, REGISTRATION_DATE)
  3 VALUES
  4 ('C004', 'Divya Kumar', 'divya@gmail.com', '9876543213', 'Velachery', TO_DATE('13-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO Customer
  2 (CUSTOMER_ID, CUSTOMER_NAME, EMAIL, PHONE, ADDRESS, REGISTRATION_DATE)
  3 VALUES
  4 ('C005', 'Karthik S', 'karthik@gmail.com', '9876543214', 'Guindy', TO_DATE('14-09-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO Customer
  2 (CUSTOMER_ID, CUSTOMER_NAME, EMAIL, PHONE, ADDRESS, REGISTRATION_DATE)
  3 VALUES
  4 ('C006', 'Meena Devi', 'meena@gmail.com', '9876543215', 'Chromepet', TO_DATE('15-09-2026','DD-MM-YYYY'));

1 row created.
```

7. Update:

```
SQL> UPDATE Customer
  2 SET CUSTOMER_NAME = 'Aakash'
  3 WHERE CUSTOMER_ID = 'C002';

1 row updated.

SQL> UPDATE Customer
  2 SET EMAIL = 'aakash35@gmail.com'
  3 WHERE CUSTOMER_ID = 'C002';

1 row updated.
```

8. Delete:

```
SQL> DELETE FROM Customer
      2 WHERE CUSTOMER_ID = 'C006';

1 row deleted.
```

After insertion, Updation and Deletion:

ADDRESS			

REGISTRAT			

C001	Arun Kumar	arun@gmail.com	9876543210
Chennai			
10-SEP-26			
C002	Aakash	aakash35@gmail.com	9876543211
Tambaram			
11-SEP-26			
CUSTO	CUSTOMER_NAME	EMAIL	PHONE

ADDRESS			

REGISTRAT			

C003	Rahul Raj	rahul@gmail.com	9876543212
Chengalpattu			
12-SEP-26			
C004	Divya Kumar	divya@gmail.com	9876543213
Velachery			
CUSTO	CUSTOMER_NAME	EMAIL	PHONE

ADDRESS			

REGISTRAT			

13-SEP-26			
C005	Karthik S	karthik@gmail.com	9876543214
Guindy			
14-SEP-26			